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Digital twin-based multi-objective parameter identification for a cable-stayed bridge

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ABSTRACT

Accurate parameter identification is essential for ensuring the structural safety and operational performance of long-span bridges. This paper presents a digital twin-based multi-objective optimization framework for identifying structural parameters of a cable-stayed bridge by matching its frequencies, mode shapes, and deflections. The framework integrates three techniques: (1) adaptive reference point relocation to maintain solution diversity across irregular Pareto fronts; (2) sensitivity-informed variation operators that concentrate exploration on parameters with significant influence; and (3) hybrid local search incorporating quasi-Newton refinement for accelerated convergence. The framework was evaluated through implementation on a case study of a 580-meter-long cable-stayed bridge. Experimental results demonstrate up to a 52% reduction in parameter identification error and 25–35% reductions in total runtime compared with the Non-dominated Sorting Genetic Algorithm II (NSGA-II) and standard NSGA-III. Robustness analysis confirms the framework's reliability with errors consistently below 3%. This research contributes to advancing bridge condition assessment capabilities, thereby enhancing structural monitoring and optimizing maintenance strategies for critical infrastructure.

1. Introduction

Cable-stayed bridges constitute a prominent category of long-span structures acclaimed for their superior structural efficiency, architectural aesthetics, and construction adaptability [1,2]. In practice, reliable bridge condition assessment is essential for ensuring operational safety and supporting risk-informed maintenance, because long-span cable--stayed systems are highly sensitive to localized deterioration and stiffness/force redistribution in safety-critical components. Nevertheless, the inherent complexities of these structures, such as geometric nonlinearities and material property uncertainties, pose considerable challenges for accurate condition assessment. In long-term service, key components such as girders, stay cables, pylons, and anchorage systems inevitably deteriorate due to fatigue damage [3], material aging, and environmental corrosion, which can reduce structural performance and trigger safety risks [4]. Since condition assessment fundamentally relies on comparing measured and predicted structural responses, establishing reliable Finite Element (FE) models and conducting effective parameter identification are critical but technically demanding tasks for integrity assurance and maintenance optimization [5–7]. Consequently, the fidelity of bridge FE models governs the precision of structural response analysis and prediction capabilities [8,9].

The mechanical responses of cable-stayed bridges are governed by numerous interdependent parameters, such as material properties, stayed-cable pretension forces, tower stiffness, and bearing support conditions [7,10]. Conventional parameter identification methods have often relied on single-objective optimization frameworks [11], wherein individual response metrics such as stayed-cable forces [12–14] or bridge deflections [15] serve as optimization targets. Although computationally tractable, these approaches exhibit a fundamental deficiency: optimization of individual response parameters fails to ensure consistency across other critical structural behaviors, resulting in parameter estimates with limited adaptability and reliability.

Moreover, when performance degradation evolves over time and

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different components deteriorate asynchronously, matching a single response metric may further amplify non-uniqueness and reduce the transferability of identified parameters across different operational states. To overcome the above limitations, researchers have pursued multi-objective optimization methods to enable comprehensive parameter identification strategies. Huang et al. [16,17] developed a bi-objective optimization framework incorporating both cable pretension forces and tower-deck deflections, whereas Peng et al. [18] formulated multiple objective functions encompassing static displacements and modal frequencies. These multi-objective approaches have demonstrated enhanced reliability through simultaneous consideration of multiple influential parameters [19–21]. However, they inevitably confront the curse of dimensionality, wherein exponentially increasing computational demands and conflicting objective functions significantly impede convergence efficiency and parameter estimation robustness.

In response to these computational bottlenecks, data-driven methods have been introduced as practical alternatives. Wu et al. [22] developed FE-based Kriging surrogate models for parameter identification, Qiu et al. [23] proposed two-phase model updating strategies integrating multi-objective optimization with elimination and choice expressing reality III (ELECTRE-III) decision-making frameworks, and Xu et al. [24] implemented enhanced particle swarm optimization algorithms for vibration control objectives. Despite these advances, computational efficiency improvements come at the cost of vulnerability to measurement noises and uncertainties that are ubiquitous in real-world bridge monitoring systems [25–27].

Collectively, these evolutionary developments reveal three persistent fundamental challenges: (1) high-dimensional parameter spaces (often exceeding 100 parameters) creating computational inefficiency and overfitting risks; (2) irregular, non-convex Pareto fronts from simultaneous multi-objective matching requirements that conventional algorithms struggle to navigate; and (3) measurement uncertainties in field monitoring data that significantly compromise identification accuracy under realistic conditions.

Addressing these challenges requires innovative optimization frameworks that transcend conventional limitations. Recent research reveals promising pathways via integrating reference-point based mechanisms [28,29] for maintaining solution diversity across irregular Pareto fronts, sensitivity analysis techniques [30] for efficient high-dimensional exploration, and hybrid optimization approaches [31,32] for enhanced robustness. While conventional algorithms such as the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [33,34] and Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D) [35] exhibit limitations in high-dimensional scenarios [36], the emergence of NSGA-III [37] with its reference-point mechanism represents a significant advancement for multi-objective optimization tasks. This transformative potential is evidenced by successful applications across diverse engineering domains: Hernandez et al. [31] achieved superior power transformer optimization through hybrid methods; Zhang et al. [32] overcame complex hydraulic parameter constraints; Sun et al. [30] integrated sensitivity analysis for high-dimensional motor optimization; and Li et al. [38] demonstrated advantages in structural bridge optimization compared with conventional algorithms [39].

The synergistic integration of these advanced techniques offers unprecedented capabilities: reference-point mechanisms employed in NSGA-III [28,29,37] ensure uniform exploration of non-convex Pareto fronts; sensitivity-informed approaches [30] conquer dimensionality curses by focusing on influential parameters while maintaining system fidelity; hybrid local-global strategies [31,32] provide robustness under realistic noise conditions through combined exploration and refinement. However, integrating these techniques into a cohesive framework specifically tailored for cable-stayed bridge parameter identification remains an open challenge.

As discussed above, this paper presents a digital twin-enabled multi-objective optimization framework that integrates a four-stage pipeline to tackle both computational complexity and measurement uncertainties in bridge parameter identification. As shown in Fig. 1, the framework consists of: (1) problem definition and setup with physics-constrained digital twin modelling; (2) sensitivity analysis for dimensionality reduction; (3) an enhanced NSGA-III algorithm with adaptive reference-point management and sensitivity-informed variation operators; and (4) multi-objective optimization followed by robustness evaluation under measurement noise. This work represents the first integrated solution that systematically addresses both complexity and uncertainty challenges in large-scale bridge parameter identification, providing a robust and computationally efficient approach essential for practical engineering applications and advancing the state-of-the-art in structural health monitoring and digital twin technologies.

2. Methods

This section presents the methodological framework for digital twin-enabled multi-objective optimization in bridge parameter identification. The methods comprise problem definition, multi-objective optimization formulation with constraint systems and sensitivity analysis, and an improved NSGA-III algorithm featuring adaptive mechanisms for enhanced convergence and robustness. Each component addresses computational complexity and measurement uncertainty in large-scale structural parameter identification.

2.1. Problem definition

Let $\theta = [\theta_1, \theta_2, \ldots, \theta_n]^T \in \Theta \subset \mathbb{R}^n$ denote the vector of parameters to be identified, where the feasible domain Θ is determined by lower and upper bounds $\theta_K^{\min} \le \theta_K \le \theta_K^{\max}$, ensuring physical plausibility and numerical stability according to design specifications, material tolerances, and field calibrations. The identification problem encompasses 156 parameters organized into four categories (Table 1): (1) pretension forces for 144 stay cables grouped into 36 symmetric sets; (2) deck material properties including elastic and shear moduli; (3) pylon structural parameters comprising elastic moduli and moment-of-inertia correction factors; and (4) modal damping ratios for the first five modes. Given the high-dimensional nature of this optimization task, sensitivity analysis is employed to identify the most influential parameters for subsequent multi-objective optimization, thereby maintaining solution accuracy while achieving computational efficiency. At the element level, the digital twin contains 156 independent parameters (cable forces, sectional stiffnesses, and damping ratios). For parameter identification, these are first grouped into 48 design variables as listed in Table 1 (e.g., 144 cables consolidated into 36 symmetric groups), and then further reduced to 16 influential parameters via Sobol sensitivity screening as described in Section 2.2.

2.2. Multi-objective optimization

The multi-objective optimization framework simultaneously addresses modal frequency matching, mode shape correlation, and static deflection accuracy without a priori preference weighting [40]. The Pareto-optimal set P^* represents the complete trade-off frontier where no objective can be improved without degrading others, as defined in Eq. (1):

$$P^* = \left\{ \theta \in \Theta \middle| \nexists \theta' \in \Theta : F_i^*(\theta') \le F_i^*(\theta) \forall i, F_j^*(\theta') < F_j^*(\theta) \text{for some } j \right\} \quad (1)$$

where $\theta \in \Theta$ is the parameter vector and $fi(\theta)(i = 1, 2, 3)$ are the objective functions.

2.2.1. Objective function

Three objective functions are used to quantify the discrepancy between numerical predictions and experimental measurements [41], as follows:

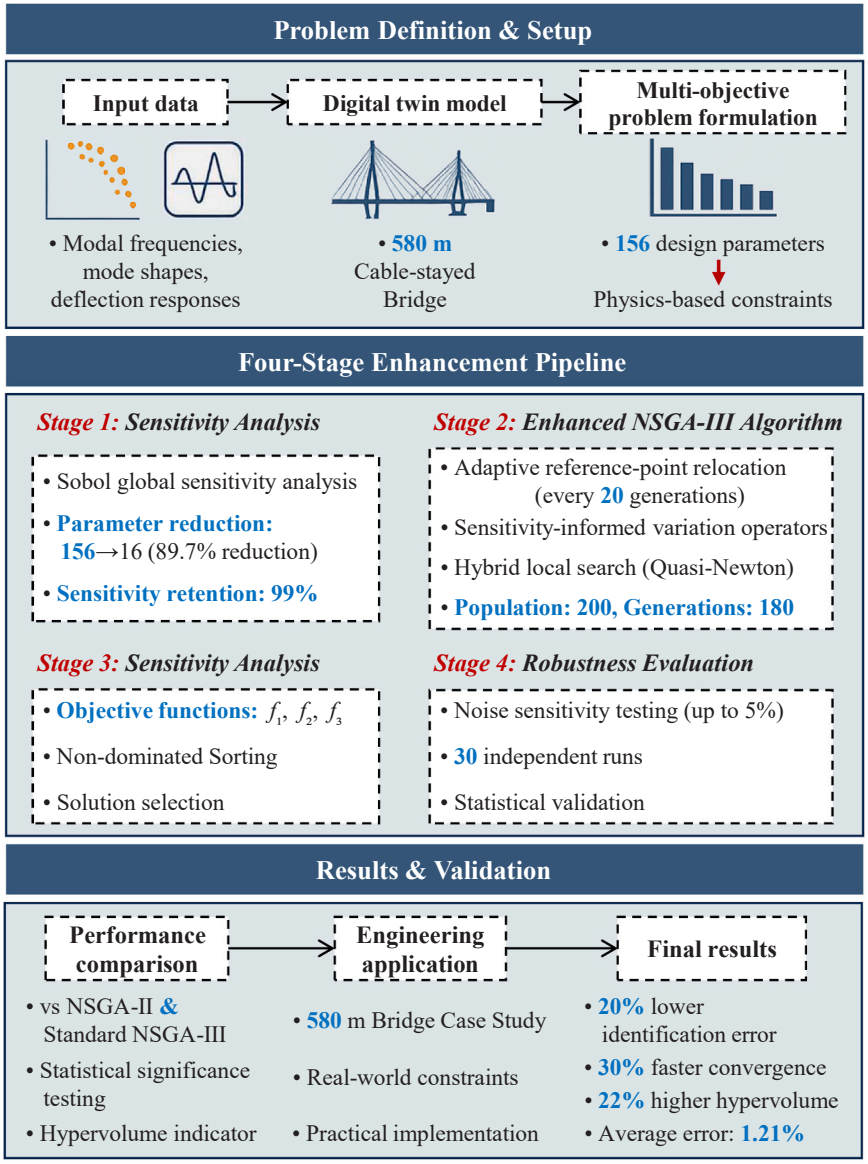


Fig. 1. Enhanced NSGA-III framework workflow for bridge parameter identification.

Normalized modal frequency error.

For the first N_m modes, the normalized modal frequency error $f_1(\theta)$ is formulated as:

$$f_1(\theta) = \frac{1}{N_m}\sum_{i=1}^{N_m}\frac{|\omega_i^{calc}(\theta) - \omega_i^{meas}|}{\omega_i^{meas}} \tag{2}$$

where $\omega_i^{calc}(\theta)$ and ω_i^{meas} denote the i-th natural frequency calculated by an FE model under parameter vector θ and the measured i-th natural frequency, respectively. Minimizing $f_1(\theta)$ is equivalent to forcing the natural frequencies predicted by the FE model to match the measurements.

Modal assurance criterion (MAC).

The MAC for the i-th mode is defined as:

$$MAC_i(\theta) = \frac{\left|\phi_i^{calc}(\theta)^T \phi_i^{meas}\right|^2}{\left\|\phi_i^{calc}(\theta)\right\|_2^2 \left\|\phi_i^{meas}\right\|_2^2} \tag{3}$$

where $\phi_i^{calc}(\theta) \in \mathbb{R}^d$ is the normalized i-th mode shape computed using the FE model under parameter vector θ and $\phi_i^{meas} \in \mathbb{R}^d$ is the corresponding measured mode shape.

The MAC complement is used to include mode shape matching as a minimization objective:

$$f_2(\theta) = \frac{1}{N_m}\sum_{i=1}^{N_m}[1 - MAC_i(\theta)] \tag{4}$$

where N_m is the total number of modes considered and $f_2(\theta)$ is the average modal-shape mismatch. Minimizing $f_2(\theta)$ maximizes mode shape correlation, with $MAC_i = 1$ indicating perfect correlation and $MAC_i = 0$ indicating orthogonality.

Normalized static deflection error.

Under M static load cases (e.g., uniform live load, asymmetric truck load), the normalized static-deflection error $f_3(\theta)$ is defined as:

$$f_3(\theta) = \frac{1}{M}\sum_{j=1}^{M}\frac{\left\|u_j^{calc}(\theta) - u_j^{meas}\right\|_2}{\left\|u_j^{meas}\right\|_2} \tag{5}$$

where $u_j^{calc}(\theta) \in \mathbb{R}^d$ is the displacement vector at sensor locations computed by the numerical model under the j-th load case, and $u_j^{meas} \in \mathbb{R}^d$ is the corresponding experimentally measured displacement

Table 1
Parameters for identification: categories, symbols, physical meanings, units, and bounds.

Category	Symbol	Physical meaning	Unit	Lower bound	Upper bound	Count
Cable	T_1~T_{36}	Pretension in stayed cables	kN	2000	5000	36
Deck	E_d	Elastic modulus of deck steel	GPa	195	230	1
	G_d	Shear modulus of deck steel	GPa	75	90	1
Pylon	E_{t1}	Elastic modulus of concrete in Tower 1	GPa	27	33	1
	E_{t2}	Elastic modulus of concrete in Tower 2	GPa	27	33	1
	I_1~I_3	Sectional moment-of-inertia correction factors	-	0.8	1.2	3
Damping	ξ_1~ξ_5	Damping ratios (the first five modes)	%	0.2	2.0	5

vector. Minimizing $f_3(\theta)$ ensures static response consistency. By simultaneously minimizing $f_1(\theta)$, $f_2(\theta)$, and $f_3(\theta)$, the optimization algorithm calibrates parameter vector θ so that the FE model reproduces both the dynamic and static characteristics observed in experimental data.

2.2.2. Constraint system formulation

To ensure physical feasibility and numerical stability, three constraint classes are imposed on the parameter vector θ as shown in Fig. 2: bound constraints, equilibrium-residual constraints, and modal-stability constraints. Bound constraints restrict parameters to physically credible ranges based on design codes and engineering practice. For instance, steel elastic modulus is limited to 195–230 GPa, ensuring material properties remain within realistic bounds:

$$\theta_K^{\min} \le \theta_K \le \theta_K^{\max}, k = 1, 2, \ldots n \tag{6}$$

where $\theta^{\min}$ and $\theta^{\max}$ are vectors of lower and upper bounds for each parameter component, respectively. Equilibrium-residual constraints require finite element equilibrium between internal forces and applied loads. The residual vector $\boldsymbol{R}(\theta) = \boldsymbol{K}(\theta)\cdot\boldsymbol{u}_s - \boldsymbol{F}_s$ must satisfy:

$$\| \boldsymbol{R}(\theta)\|_2 \le \varepsilon_R \tag{7}$$

where $\boldsymbol{K}(\theta)$ is the parameter-dependent stiffness matrix, u_s is the static displacement vector, $\boldsymbol{F}_s$ is the applied load vector, $||\cdot||_2$ denotes the Euclidean norm, and ε_R is the relative equilibrium tolerance (typically 10^{-6}). Modal-stability constraints ensure positive eigenvalues to avoid numerical instabilities and the mathematical expression is given in Eq. (8):

$$[K(\theta) - \lambda_i M]\varphi_i = 0 \tag{8}$$

where $\boldsymbol{M}$ is the mass matrix, φ_i is the i-th mode shape vector, and λ_i relates the eigenvalue to the i-th natural frequency ω_i.

The constrained problem is converted to unconstrained form via quadratic penalties:

$$F_i^*(\theta) = F_i(\theta) + \omega_b \sum_{k=1}^{n} [\max(0, \theta_k^{\min} - \theta_k]^2 + \omega_b \sum_{k=1}^{n} [\max(0, \theta_k - \theta_k^{\min}]^2$$

$$+ \omega_e \|R(\theta)\|_2^2 \tag{9}$$

where $i = 1, 2, 3$ and penalty weights ω_b, $\omega_e \gg 1$ heavily penalize constraint violations.

2.2.3. Parameter sensitivity analysis

High-dimensional optimization faces computational challenges and overfitting risks. Parameter sensitivity analysis quantifies each parameter's importance, enabling dimensionality reduction while preserving solution quality.

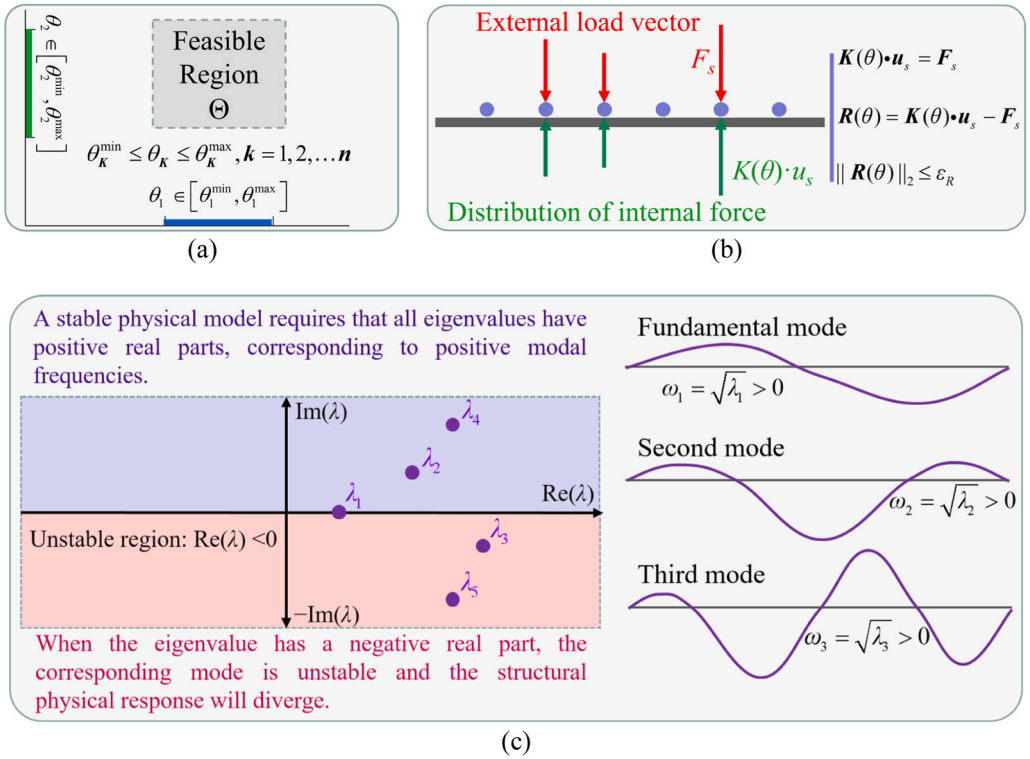


Fig. 2. Constraint formulations: (a) parameter bound constraints; (b) equilibrium constraints; and (c) modal stability constraints.

Sobol variance-based sensitivity analysis quantifies the influence of each parameter θ_k on the penalized objective vector $F^*(\theta) = [F_1^*(\theta), F_2^*(\theta), F_3^*(\theta)]$. The total sensitivity index S_T^K measures the fraction of output variance attributable to θ_k (including interactions), and lies between 0 and 1. Using Quasi-Monte Carlo Sobol sequences with sample size N_s per parameter, first- and total-order indices are computed via the Saltelli scheme [42].

Parameter screening fixes parameter k at its nominal design value when:

$$S_T^K < \delta_S \tag{10}$$

where $\delta_S = 0.01$ is the sensitivity threshold. This yields a reduced decision space $\Theta_{reduced} \subset \mathbb{R}^{\{n^r\}}, n^r \ll n$.

Objective normalization addresses scale disparities that can bias multi-objective operators. Each penalized objective is normalized to [0,1] using preliminary sample extrema:

$$\widehat{F}_i(\theta) = \frac{F_i^*(\theta) - \min_{\zeta \in \Theta_{sample}} F_i^*(\zeta)}{\max_{\zeta \in \Theta_{sample}} F_i^*(\zeta) - \min_{\zeta \in \Theta_{sample}} F_i^*(\zeta)} \tag{11}$$

where $\min_{\zeta \in \Theta_{sample}} F_i^*(\zeta)$ and $\max_{\zeta \in \Theta_{sample}} F_i^*(\zeta)$ are the minimum and maximum values of the i-th objective observed during preliminary sampling.

Fig. 3 presents Sobol global sensitivity analysis quantifying parameter importance across the three optimization objectives. The analysis reveals distinct sensitivity patterns: cable pretensions (T_1, T_2) dominate modal frequency objective f_1 (sensitivity indices 0.6–0.8), tower elastic moduli (E_{t1}, E_{t2}) primarily affect mode shape objective f_2 (0.4–0.5), while deck elastic modulus (E_d) overwhelmingly influences static deflection objective f_3 (0.65).

Using sensitivity threshold $\delta_S = 0.01$, systematic screening reduces the parameter space from 156 to 16 parameters (89.7% reduction) while retaining 99% of system sensitivity, providing both computational efficiency and scientific foundation for the enhanced NSGA-III algorithm.

2.3. Improved NSGA-III algorithm

To address high-dimensional, constrained multi-objective parameter identification in large-span cable-stayed bridges, several enhancements are implemented to the standard NSGA-III framework [43]. The key innovations are: An adaptive reference-point relocation strategy to maintain diversity in irregular Pareto fronts; Sensitivity-informed variation operators that bias exploration toward the most influential parameters; A hybrid local-search module that accelerates fine-scale convergence; A brief convergence discussion under realistic modeling assumptions.

2.3.1. Baseline NSGA-III framework

NSGA-III extends NSGA-II to multi-objective problems (M≥3) by

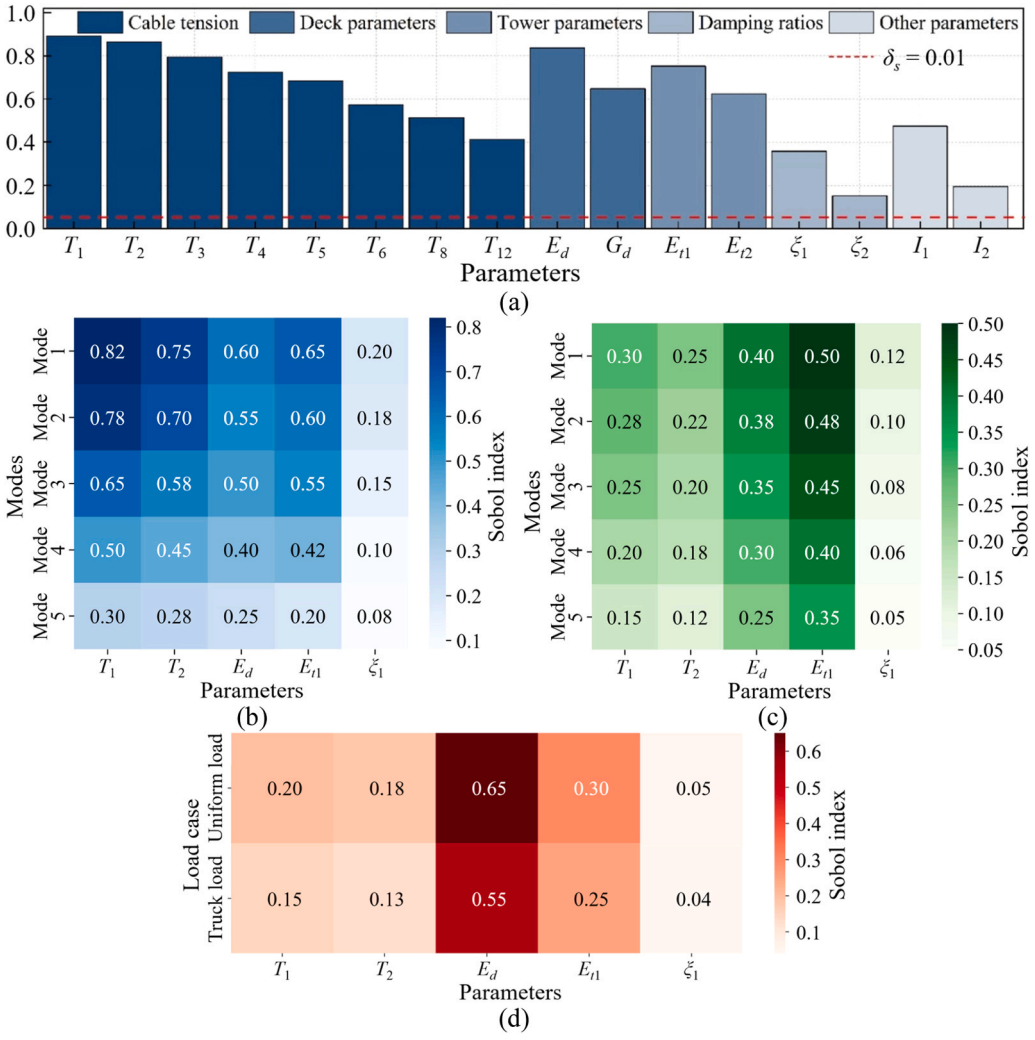


Fig. 3. Sobol global sensitivity analysis: (a) sensitivity index; (b) modal frequency sensitivity heatmap; (c) modal shape sensitivity heatmap; and (d) static deflection sensitivity heatmap.

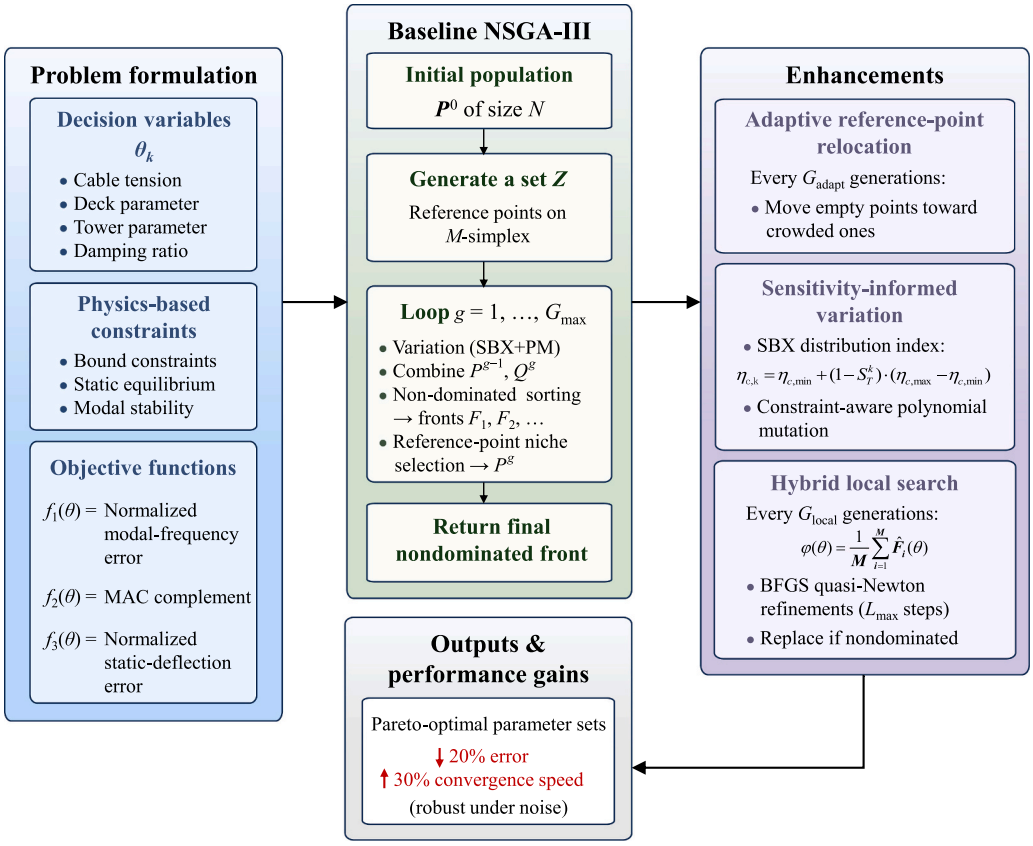


Fig. 4. Flowchart of the baseline NSGA-III algorithm.

replacing crowding-distance with a reference-point-based niche preservation. Fig. 4 illustrates the baseline algorithm workflow, which proceeds as follows:

(1) Initialization: Generate an initial population P^0 of size N by uniform sampling in decision space $\Theta_{reduced}$.
(2) Reference point generation: Construct a set $Z = \{z^1, \ldots z^H\}$ of H uniformly distributed reference points on the unit M-simplex.
(3) Evolution loop (for $g = 1, \ldots, G_{max}$):

- Variation: Apply binary tournament selection, simulated binary crossover (SBX) and polynomial mutation to P^{g-1}, producing offspring Q^g.
- Combination: $R^g = P^{g-1} \cup Q^g$.
- Non-dominated Sorting: Decompose R^g into fronts $F_1, F_2, \ldots$.
- Population Reduction: Fill new population P^g front by front. When the next front F_k cannot be entirely accommodated, apply reference-point niche selection: project individuals in F_k onto the normalized objective space; associate each with nearest reference point in Z and prefer individuals from underrepresented niches until N members are selected.

(4) Termination: Return nondominated set in P^G_{max}

While effective for multi-objective optimization, the vanilla NSGA-III may lose diversity when Pareto fronts are non-convex or exhibit uneven density, and applies generic variation operators that ignore problem-specific parameter sensitivity.

2.3.2. *Reference point adaptation*

Fixed reference grids can cause solution clustering in dense Pareto front regions and gaps in sparsely populated areas, particularly for non-convex trade-off surfaces common in cable-stayed bridge calibration. To address this limitation, reference points are periodically adjusted to better track the evolving front geometry.

Table 2 details the adaptive reference-point adjustment mechanism parameters for the improved NSGA-III. For three-dimensional objective

Table 2
Reference point adaptation parameters and settings.

Parameter	Symbol	Value	Description
Reference-point generation dimension	M	3	Number of objectives (three-dimensional objective space)
Simplex division count	p	12	Division parameter in the Das & Dennis method
Total number of reference points	H	220	Interior + boundary points
Number of interior reference points	H_1	91	First-layer (interior) reference-point count
Number of boundary reference points	H_2	129	Second-layer (boundary) reference-point count
Adaptation interval	G_{adapt}	20	Detect and adjust reference-point distribution every 20 generations
Minimum associated-solution threshold	ρ_{min}	1	Reference points with fewer associated solutions than this are deemed "empty"
Overcrowding threshold	ρ_{max}	5	Reference points with more associated solutions than this are deemed "overcrowded"
Mixing factor for relocation	α	0.5	Weight balancing old vs. new position ($\in$ [0,1])
Maximum adjustment ratio	β_{max}	0.3	Maximum fraction of reference points to move per adaptation to avoid overshooting
Reference-point normalization method	-	ASF	Achievement scalarizing function

space ($M = 3$), 220 reference points are generated using the Das & Dennis method (91 interior and 129 boundary points) [44]. The adaptation mechanism inspects solution associations every $G_{adapt} = 20$ generations. "Empty" reference points (associated with fewer than ρ_{min}=1 solutions) are relocated toward "crowded" points (associated with more than ρ_{max}=5 solutions) using:

$$z_j^{new} = (1-\alpha)\cdot z_j + \alpha \cdot z_{nearest} \tag{12}$$

where $\alpha = 0.5$ is the mixing factor, dynamically redistributing reference points to maximize uniform coverage across the entire Pareto front.

Fig. 5 demonstrates the advantage of adaptive reference point management. Standard NSGA-III (Fig. 5**(a)**) employs uniformly distributed reference points across the entire [0,1] × [0,1] objective space, creating misalignment with the actual Pareto front and resulting in sparse coverage in critical non-convex regions. The improved algorithm (Fig. 5**(b)**) implements dynamic relocation every 20 generations, systematically moving empty reference points toward high-concentration areas, achieving superior front coverage and eliminating redundant points in infeasible regions.

Fig. 6 provides comprehensive performance evaluation across key multi-objective metrics. The improved algorithm achieves 12% enhancement in hypervolume coverage compared with standard NSGA-III (Fig. 6**(a)**), indicating better Pareto space exploration. Diversity analysis (Fig. 6**(b)**) shows consistent superiority across all front regions, including the depression zone (0.28 vs 0.25), convex zone (0.30 vs 0.28), and overall diversity (0.36 vs 0.32). The inverted generational distance (IGD) convergence (Fig. 6**(c)**) is 17% lower than that of standard NSGA-III at termination, confirming superior convergence quality and closer proximity to the true Pareto front.

2.3.3. Sensitivity-informed variation operators

Global sensitivity analysis yields Sobol total indices $S_T^K \in [0,1], k = 1, \ldots n_r$ quantifying parameter importance. These indices are embedded into crossover and mutation operators to bias search effort toward influential parameters.

(1) SBX with adaptive distribution index

For each decision variable θ_k, the SBX distribution index is set as:

$$\eta_{c,k} = \eta_{c,\,\min} + (1 - S_T^k)\cdot(\eta_{c,\,\max} - \eta_{c,\,\min}) \tag{13}$$

where $\eta_{c,\,\min}$ and $\eta_{c,\,\max}$ define the allowable range.

This adaptive mechanism establishes an inverse relationship between parameter sensitivity and distribution index.

(2) Constraint-aware polynomial mutation

Standard polynomial mutation employs adaptive index η_m following Eq. (13). After mutating θ_k, if the result violates bounds $[\theta_k^{\min}, \theta_k^{\max}]$ or creates equilibrium residual $r(\theta) > r_{max}$, re-mutation occurs up to M_{max} attempts before clamping to the nearest feasible bound.

This targeted operator design accelerates convergence by concentrating search effort on parameters with maximum structural influence.

2.3.4. Hybrid local search strategy

Global MOEAs can stagnate near Pareto fronts due to diminishing objective gradients. A gradient-based local search is interleaved every G_{local} generations to enhance fine-scale convergence.

Scalarization: For each nondominated solution $\theta^{(p)}$, form the simple average

$$\varphi(\theta) = \frac{1}{M}\sum_{i=1}^{M} \widehat{F}_i(\theta) \tag{14}$$

where $\widehat{F}_i$ is the normalized objectives from Eq. (11).

Quasi-Newton optimization:

Approximate $\nabla\varphi(\theta^{(p)})$ via central finite differences with step size $h = 10^{-6}$.

Perform up to L_{max} Broyden-Fletcher-Goldfarb-Shanno (BFGS) iterations subject to bounds $\theta \in \Theta$.

Terminate early if $\|\nabla\varphi\| < \varepsilon_{loc} = 10^{-4}$.

Replacement: If the refined solution remains nondominated with respect to the current population, replace the original solution.

The hybrid local search mechanism provides dual advantages: elimination of Pareto front discontinuities and enhancement of trade-off resolution quality. While introducing computational overhead, this cost is offset by substantial reduction in required global evolutionary iterations, ultimately achieving superior convergence efficiency toward high-quality Pareto-optimal solutions.

3. Case Study

The proposed multi-objective parameter identification framework is validated on a high-fidelity finite element model of a 580-m span cable-stayed bridge. This section presents the bridge model configuration, synthetic measurement data, optimization setup, and quantitative results demonstrating identification accuracy, convergence speed, and Pareto front diversity. Fig. 7 illustrates the comprehensive sensor deployment strategy employed for structural response measurements.

3.1. Bridge model description

This section presents the finite element model of the cable-stayed bridge used for validation of the proposed parameter identification

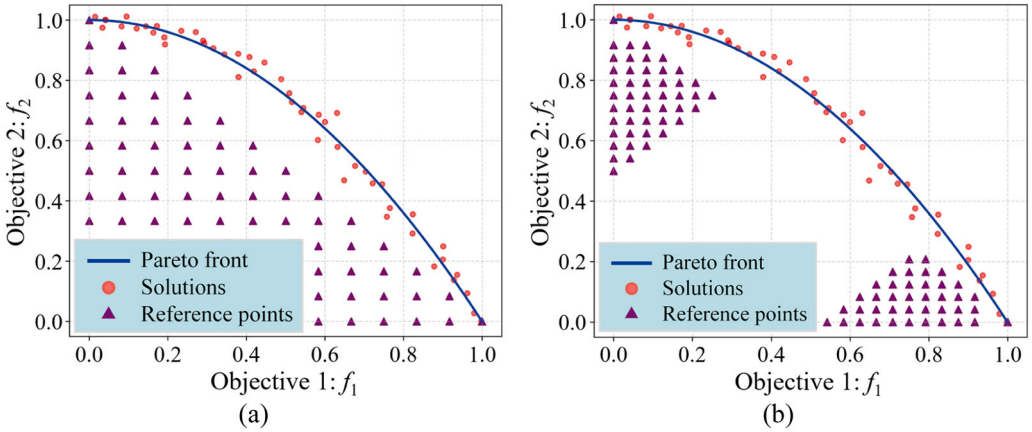


Fig. 5. Reference point adaptation: (a) standard NSGA-III with fixed reference points; (b) improved NSGA-III with adaptive reference points.

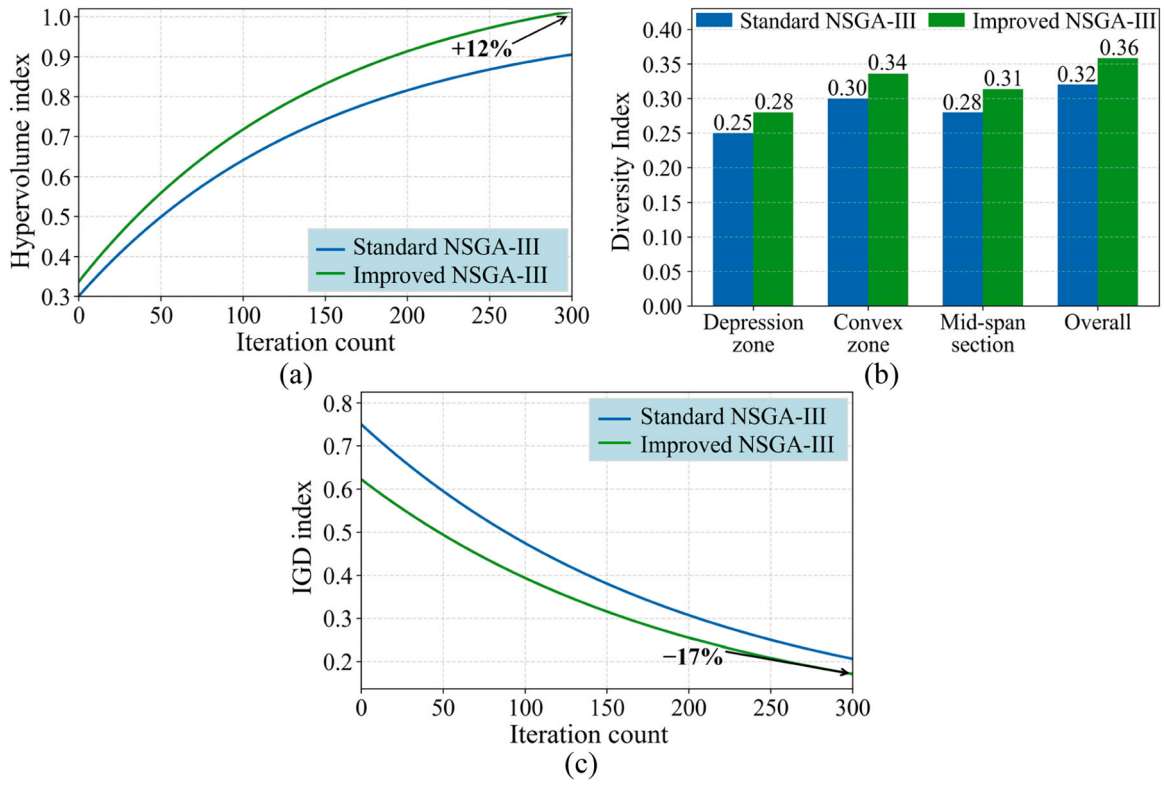


Fig. 6. Multi-metric performance comparison of standard and improved NSGA-III algorithms: (a) hypervolume convergence; (b) pareto front diversity; and (c) IGD metric convergence.

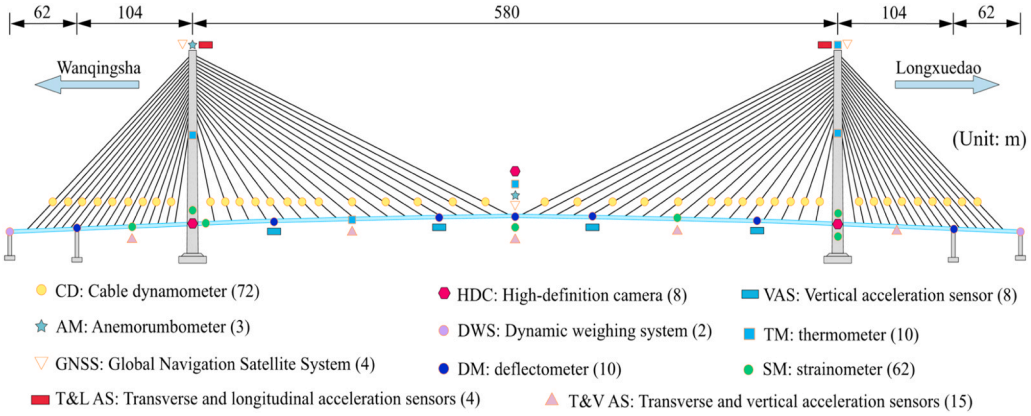


Fig. 7. Sensor deployment scheme for cable-stayed bridge monitoring.

Table 3
Finite element model properties of cable-stayed bridge.

Parameter Category	Value	Unit	Description
Deck elastic modulus, E_d	210	GPa	Orthotropic steel panels
Steel density, ρ_d	7850	kg/m^3	Orthotropic steel panels
Deck Poisson's ratio, ν_d	0.3	-	Orthotropic steel panels
Deck shear modulus, G_d	80.8	GPa	Orthotropic steel panels
Tower elastic modulus, E_t	30	GPa	C60 concrete
Tower density, ρ_t	2500	kg/m^3	C60 concrete
Tower Poisson's ratio, ν_t	0.2	-	C60 concrete
Cable elastic modulus, E_c	195	GPa	Parallel steel strands
Cable prestress range	2000–5000	kN	Identification parameter
Damping ratio range, ξ	0.2–2.0	%	Identification parameter
Modal analysis	First 5 modes	-	Primary contributors
Tower base	Fixed	-	Fully constrained
Deck ends	Elastic bearings	-	Vertical sliding allowed

framework. Table 3 details the main parameters of the cable-stayed bridge finite element model. The bridge features total length of 912 m (62 m+104 m+580 m+104 m+62 m) with 30-m deck width. Material properties include orthotropic steel deck panels with elastic modulus E_d = 210 GPa, Poisson's ratio $\nu_d = 0.3$, and shear modulus $G_d = E_d/[2(1+\nu_d)] \approx 80.8$ GPa; C60 high-performance concrete towers with elastic modulus $E_t = 30$ GPa; and high-strength parallel wire cable strands (diameter 80–160 mm, elastic modulus $E_c = 195$ GPa) with prestress forces serving as key identification parameters. The model contains approximately 12,500 elements (10,000 shell, 2500 beam) with around 150,000 ° of freedom, accounting for geometric nonlinearity under large deformations, as shown in Fig. 8. Rayleigh damping with unknown modal damping ratios is assumed. Boundary conditions include fixed tower bases and elastic bearings at deck ends, providing a high-accuracy numerical platform for parameter identification. All computational experiments were conducted on a personal workstation

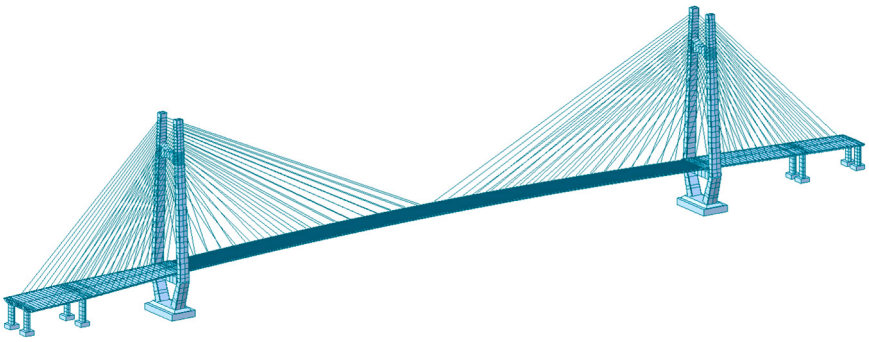


Fig. 8. The FE model.

equipped with an Intel Core i7–10700 processor (2.9 GHz, 8 cores), 128 GB DDR4 RAM, and an NVIDIA GeForce RTX 3090 Ti GPU (24 GB VRAM).

3.2. Parameter identification setup

The parameter identification framework employs sensitivity-driven parameter selection to reduce computational complexity while maintaining accuracy. Based on the cable-stayed bridge configuration, the structure features 36 pairs of stay cables symmetrically arranged on both sides of the bridge deck. Among these, nine critical cable segments are selected for identification: T_1 and T_2 represent tower-base cables anchored near the tower foundation; T_3, T_4, T_5 and T_6 denote transition-zone cables in the quarter-span region; while T_8, T_{12} and T_{15} correspond to mid-span cables experiencing maximum tension forces and load redistribution effects.

Table 4 presents comprehensive statistical analysis based on 30 independent algorithm runs, demonstrating exceptional identification performance with an average absolute relative error of 1.21%, falling well within typical engineering tolerance thresholds (±5%). All parameters exhibit coefficients of variation below 3%, indicating high algorithmic stability and reproducibility across stochastic runs. The 95% confidence intervals, contained within ±3.5% bounds, provide robust uncertainty quantification essential for engineering decision-making. The reduced parameter set retains 99% of system parametric influence while significantly enhancing computational efficiency, effectively addressing the curse of dimensionality in large-scale structural optimization.

Fig. 9 presents comprehensive performance comparison across three critical metrics. The improved NSGA-III achieves about 20% lower average objective error than standard NSGA-III (Fig. 9**(a)**) and converges 25–29% faster, reaching convergence at generation 180 compared to 240 for standard NSGA-III and 255 for NSGA-II. Hypervolume evolution (Fig. 9**(b)**) shows 25% improvement in solution diversity and Pareto front coverage. Despite 9.5% increase in per-generation computational cost (81 s vs. 74 s), the accelerated convergence yields 35% overall time savings compared to standard NSGA-III (3.2 h vs. 4.9 h) and 41% savings versus NSGA-II (3.2 h vs. 5.4 h) as shown in Fig. 9**(c)**, establishing practical viability for periodic offline updates in structural health monitoring.

Table 4
Parameter identification results and statistical validation (30 independent runs).

Parameter	True value	Mean	Std deviation	CV (%)	Relative error (%)	95% confidence interval
T_1 (kN)	3580	3632	48.6	1.34	+ 1.45	[3537, 3727]
T_2 (kN)	3750	3818	52.4	1.37	+ 1.81	[3715, 3921]
T_3 (kN)	4120	4165	45.8	1.10	+ 1.09	[4075, 4255]
T_4 (kN)	4380	4325	56.2	1.30	−1.26	[4215, 4435]
T_5 (kN)	4580	4514	62.8	1.39	−1.44	[4391, 4637]
T_6 (kN)	4650	4675	43.2	0.92	+ 0.54	[4590, 4760]
T_8 (kN)	4820	4785	51.6	1.08	−0.73	[4684, 4886]
T_{12} (kN)	5150	5095	72.4	1.42	−1.07	[4953, 5237]
T_{15} (kN)	5350	5318	84.6	1.59	−0.60	[5152, 5484]
E_d (GPa)	210.0	206.8	3.25	1.57	−1.52	[200, 213]
G_d (GPa)	80.8	81.9	1.28	1.56	+ 1.36	[79.4, 84.4]
E_{t1} (GPa)	30.0	29.5	0.42	1.42	−1.67	[28.7, 30.3]
E_{t2} (GPa)	30.0	30.4	0.38	1.25	+ 1.33	[29.7, 31.1]
I_1	1.00	0.98	0.018	1.84	−2.00	[0.94, 1.02]
ξ_1 (%)	0.8	0.79	0.021	2.66	−1.25	[0.75, 0.83]
Average	-	-	-	1.45	±1.21	±3.5%

Note: Std deviation and CV denote standard deviation and coefficient of variation, respectively, quantifying the dispersion of 30 independent runs.

4. Results and discussion

This section presents the experimental results obtained using the methodology outlined in **Section 2** and the case study configuration described in **Section 3**. Three key aspects are analyzed: (1) comparative performance evaluation among improved NSGA-III, standard NSGA-III, and NSGA-II algorithms; (2) robustness assessment under measurement noise conditions; and (3) comprehensive discussion of algorithmic mechanisms and practical implications.

4.1. Algorithm performance comparison

Fig. 10 demonstrates the superior parameter identification accuracy of the improved NSGA-III algorithm through comprehensive error analysis across critical structural parameters. For cable tension identification (Fig. 10**(a)**), nine representative cable segments (T_1, T_2, T_3, T_4, T_5, T_6, T_8, T_{12}, T_{15}) spanning from tower-base to mid-span regions are evaluated.

The improved algorithm consistently achieves the lowest identification errors, with an average error of 1.4% compared to 3.3% for standard NSGA-III and 4.4% for NSGA-II, which represents accuracy improvements of 58% and 68%, respectively. On the other hand, material parameter identification results (Fig. 10**(b)**) exhibit even more pronounced advantages. For deck elastic modulus E_d, the improved algorithm achieves 1.2% error compared to 2.3% (standard NSGA-III) and 3.5% (NSGA-II), representing 48% and 66% error reductions. Similar superiority is observed for shear modulus G_d (0.9% vs. 1.8% vs. 2.5%), tower elastic modulus E_{t1} (1.0% vs. 2.0% vs. 2.8%), and damping ratio ξ_1 (0.5% vs. 0.8% vs. 1.2%). Notably, the improved algorithm maintains maximum error below 1.2% across all material parameters, whereas competing algorithms exhibit higher variability with peak errors approaching 3.5%. This precision is important because elastic moduli and damping parameters directly govern serviceability predictions and fatigue life assessments.

The consistently superior performance validates the synergistic effectiveness of three algorithmic enhancements: (1) adaptive reference point relocation maintains solution diversity across irregular Pareto regions, preventing premature convergence; (2) sensitivity-informed operators concentrate computational effort on influential parameters

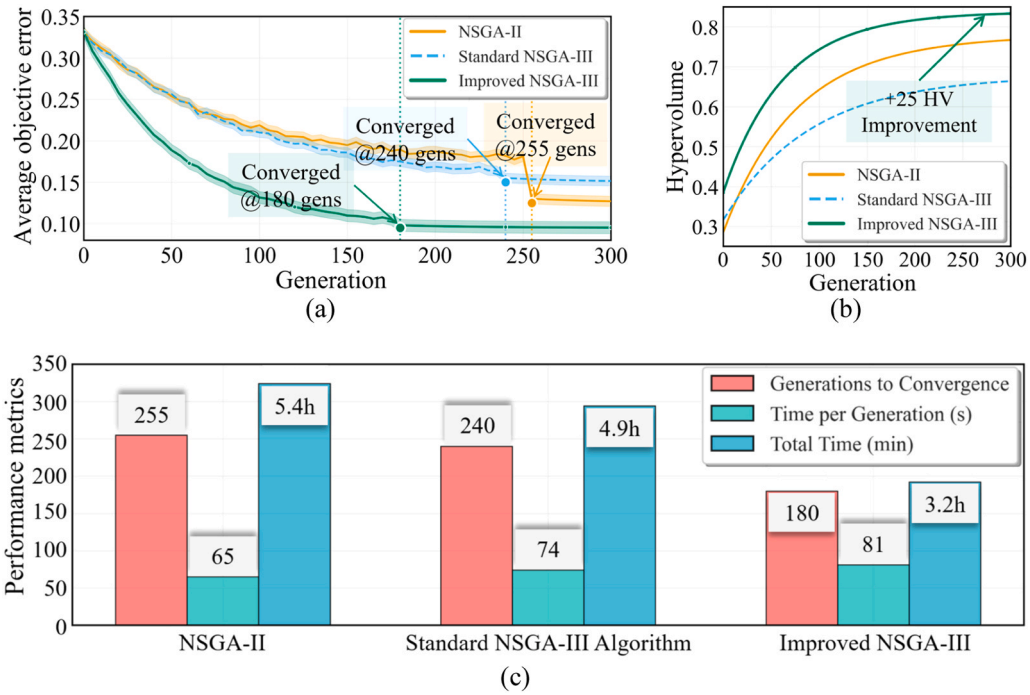


Fig. 9. Algorithm convergence and efficiency comparison for parameter identification of the cable-stayed bridge: (a) convergence curves; (b) hypervolume trajectories; and (c) breakdown of computational efficiency metrics.

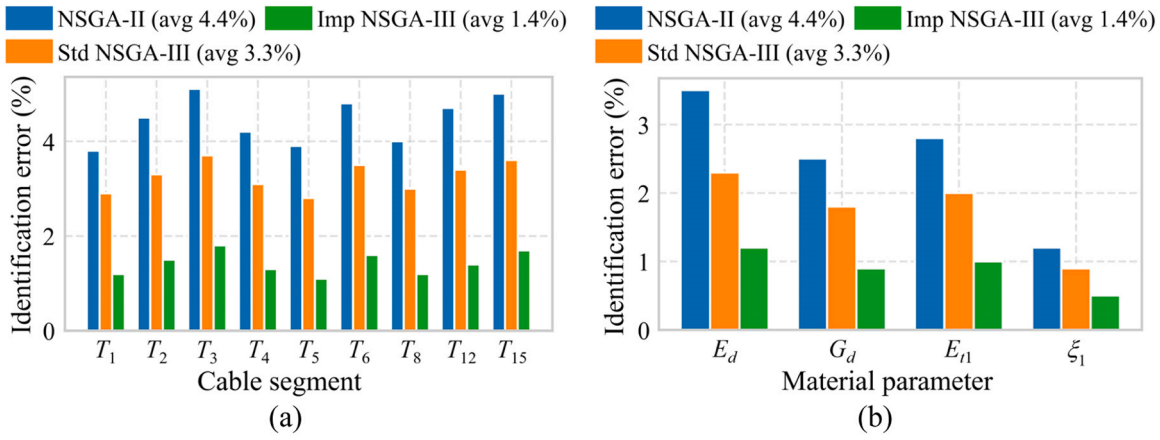


Fig. 10. Comparative identification accuracy of three optimization algorithms: (a) cable tension identification errors; and (b) material parameter identification errors.

identified through global sensitivity analysis; and (3) hybrid local search refines solutions near Pareto optimality, eliminating residual errors. These coordinated improvements establish the proposed framework as highly effective for high-fidelity digital twin construction of cable-stayed bridges, where parameter accuracy directly impacts structural safety assessment and long-term health monitoring reliability.

4.2. Robustness assessment under measurement noise

To evaluate algorithm robustness under realistic field conditions, comprehensive noise sensitivity analysis was conducted across three progressively challenging scenarios: S1 (low noise, 1% modal frequency error and 0.5 mm deflection noise), S2 (baseline, 2% frequency and 1 mm deflection noise), and S3 (high noise, 5% frequency and 2 mm deflection noise). For each scenario, 30 independent algorithm runs were performed to assess: (1) average normalized objective at termination; (2) mean relative parameter identification error across 16 reduced parameters; (3) generations required for convergence (defined as no improvement in objective for 20 consecutive generations); and (4) hypervolume of the final Pareto front.

Table 5 demonstrates exceptional algorithm robustness across noise scenarios. Under low noise (S1), the improved NSGA-III achieves overall average error of 0.9%, representing a 55% reduction compared to standard NSGA-III (2.0%). At baseline noise (S2), average error increases moderately to 1.3% (52% improvement), while under high noise (S3) it remains at 2.8% (39% improvement). Parameter category analysis under S2 shows consistent performance: cable tensions (1.4% error), deck properties (1.4%), and tower parameters (1.1%). Convergence acceleration of 25–29% and hypervolume improvements of 18–24% are maintained across scenarios, while computational time reductions of 30–37% offset the 9.5% per-generation overhead. The noise sensitivity coefficient of 0.75 (vs. 0.85 for standard NSGA-III, 0.90 for NSGA-II) confirms graceful performance degradation, validating practical applicability for real-world structural health monitoring.

Table 5
Noise sensitivity analysis results across measurement scenarios.

Evaluation metric	Noise level	NSGA-II	Standard NSGA-III	Improved NSGA-III	Improvement*
Cable tension error (%)	S1	2.8	2.1	0.8	−62%
	S2	4.3	3.2	1.4	−56%
	S3	6.8	5.2	2.6	−50%
Deck parameter error (%)	S1	3.0	2.0	1.0	−50%
	S2	3.8	2.8	1.4	−50%
	S3	6.0	4.5	3.0	−33%
Tower parameter error (%)	S1	3.2	1.8	1.1	−39%
	S2	3.0	2.1	1.1	−48%
	S3	5.2	3.8	2.5	−34%
Overall error (%)	S1	3.0	2.0	0.9	−55%
	S2	3.8	2.7	1.3	−52%
	S3	6.0	4.6	2.8	−39%
Convergence (generations)	S1	220	210	150	−29%
	S2	255	240	180	−25%
	S3	300 +	300 +	240	−20%
Hypervolume index	S1	0.312	0.348	0.412	+ 18%
	S2	0.285	0.316	0.385	+ 22%
	S3	0.218	0.265	0.328	+ 24%
Computational time (h)	S1	4.0	4.3	2.7	−37%
	S2	4.6	4.9	3.2	−35%
	S3	5.4 +	6.1 +	4.3	−30%
Noise sensitivity coefficient	-	0.90	0.85	0.75	−12%

Note: Improvement* values are computed with respect to the standard NSGA-III results (negative values indicate reductions, positive values indicate increases).

Fig. 11 illustrates the relationship between measurement noise and identification accuracy. The improved NSGA-III maintains superior performance across both frequency (Fig. 11**(a)**) and displacement noise scenarios (Fig. 11**(b)**), with accuracy degrading gracefully from 99.5% under low noise to 96.1% under high noise. The noise sensitivity coefficient of 0.75 significantly outperforms standard NSGA-III (0.85) and NSGA-II (0.90), confirming exceptional robustness against measurement uncertainties typical in field applications.

Fig. 12**(a)** presents average identification errors across parameter categories through bar chart comparison, showing the improved algorithm consistently achieves lowest errors for cable tensions (0.8–2.6%), deck parameters (1.0–3.0%), and tower parameters (1.1–2.5%), with performance advantages becoming more pronounced under higher noise conditions. Fig. 12**(b)** provides three-dimensional surface visualization demonstrating complex interplay between noise levels, convergence generations, and identification accuracy, confirming faster convergence (150–240 generations) with consistently lower errors across the entire noise spectrum.

Fig. 13 provides comprehensive validation of identified parameters against synthetic measurements, showing excellent agreement across multiple response domains: modal frequencies with correlation coefficients exceeding 0.98 (Fig. 13**(a)**), mode shape correlations (MAC values) above 0.95 (Fig. 13**(b)**), and static responses under uniform loading (5 kN/m, Fig. 13**(c)**) and concentrated loading (200 kN, Fig. 13**(d)**) with all residuals within ±2% engineering accuracy thresholds. These results demonstrate that optimized parameters successfully reproduce target structural behavior across domains essential for reliable digital twin applications.

4.3. *Mechanistic insights and practical implications*

The enhanced NSGA-III algorithm demonstrates that carefully tailored evolutionary strategies can effectively address high-

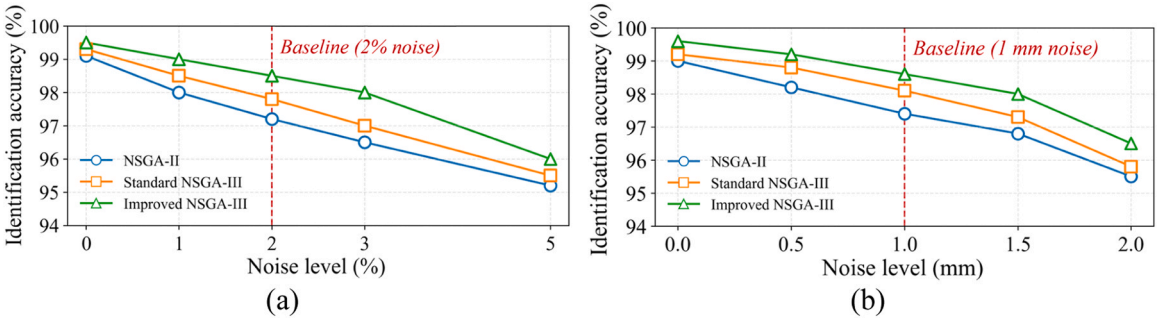


Fig. 11. Effect of measurement noise on parameter identification accuracy: (a) identification accuracy versus modal frequency noise level; (b) identification accuracy versus displacement measurement noise.

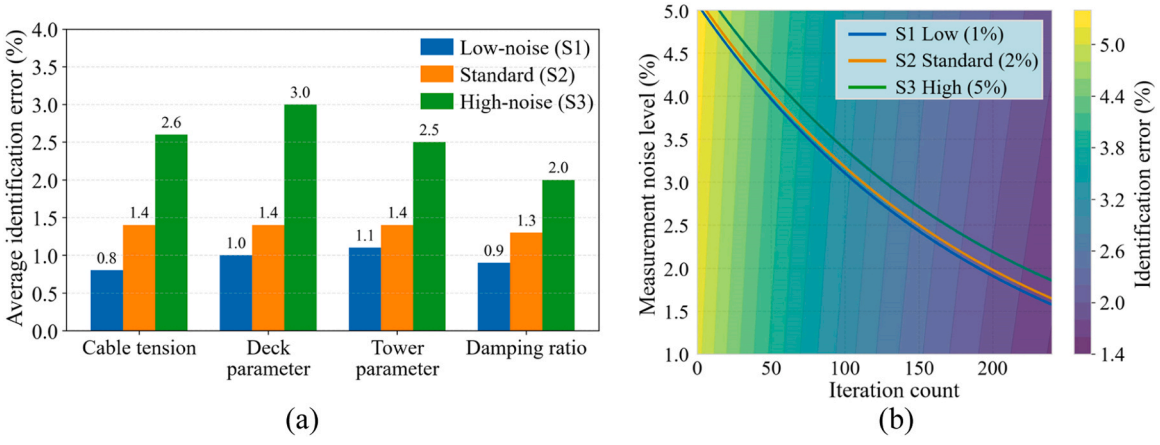


Fig. 12. Comprehensive error analysis under varying noise scenarios: (a) average parameter identification errors categorized by structural component (cables, deck, towers) across three noise levels; (b) three-dimensional noise-convergence-accuracy relationships.

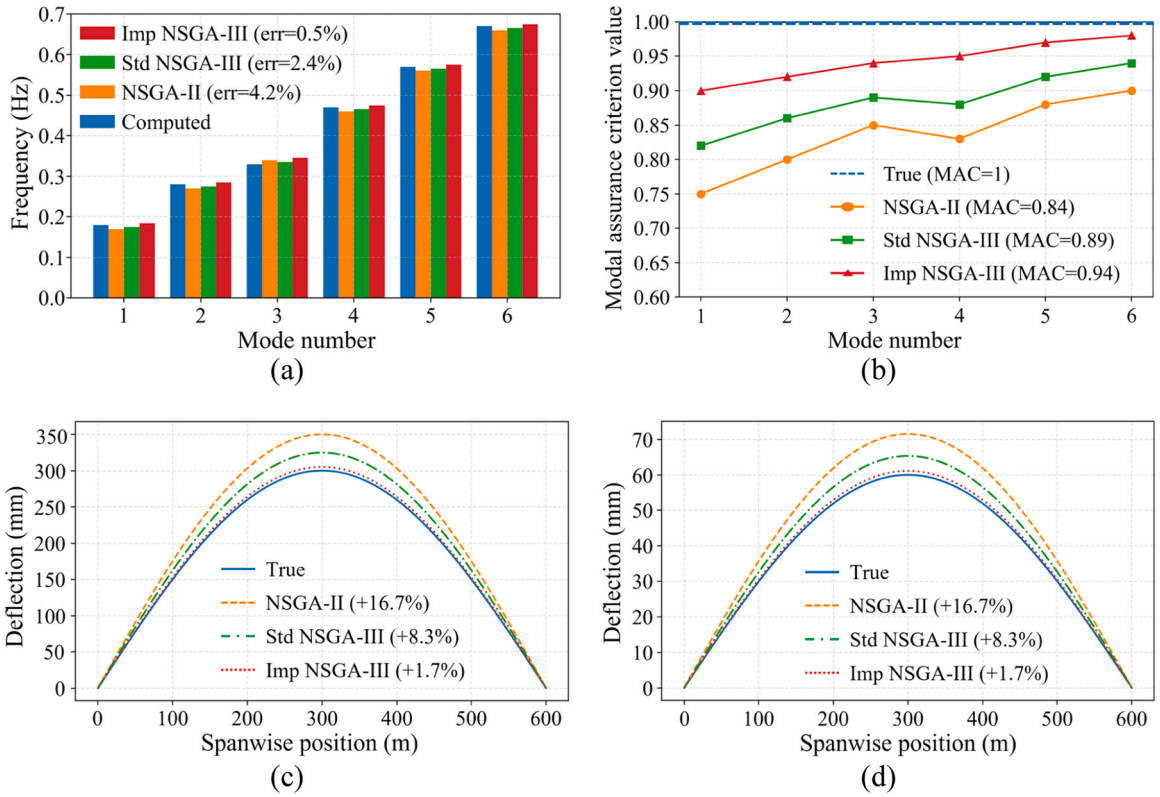


Fig. 13. Validation of identified parameters against synthetic measurement data: (a) modal frequency identification; (b) modal assurance criterion value; (c) mid-span deflection under uniform distributed load (5 kN/m); (d) deflection profile under concentrated point load (200 kN at mid-span).

dimensional, noisy, multi-objective calibration challenges inherent in cable-stayed bridge digital twin construction. The integrated framework combines three synergistic innovations that collectively improve both identification accuracy and computational efficiency.

Sensitivity-driven dimensionality reduction proves essential for tractable optimization in high-parameter spaces. By applying Sobol global sensitivity analysis to screen 156 candidate parameters and retain only the 16 most influential variables (capturing 99% of cumulative system variance), the framework concentrates computational effort on regions where structural response sensitivity is maximized. This strategic reduction mitigates overfitting risks, enhances convergence reliability, and enables practical deployment in structural health monitoring systems. Further embedding sensitivity indices into SBX crossover and polynomial mutation operators biases evolutionary exploration toward high-impact parameters, thereby accelerating error reduction across modal frequencies, mode shapes, and static deflection responses.

Adaptive Pareto front management maintains solution diversity essential for revealing complete trade-off spectra among competing calibration objectives. The dynamic reference point relocation mechanism, which redistributes underutilized reference points toward overcrowded regions every 20 generations, achieves 22% higher hypervolume compared to fixed reference grid strategies under baseline noise conditions (scenario S2, Table 5) and effectively prevents solution clustering in locally convex regions. By integrating hybrid local search that intermittently refines non-dominated solutions via BFGS iterations, this dual-scale optimization strategy reduces required generations by approximately 25% (scenario S2, Table 5) while eliminating gaps in the Pareto front without sacrificing global exploration capability.

Robustness under measurement uncertainty is confirmed through progressively challenging noise scenarios spanning 1–5% modal frequency disturbances and 0.5–2 mm deflection noise (scenarios S1–S3, Table 5). Even under the highest noise level (S3: 5% frequency error, 2 mm deflection noise), the improved NSGA-III maintains average parameter errors at 2.8%, representing 39% and 53% improvements relative to standard NSGA-III and NSGA-II, respectively. Although adaptive mechanisms introduce modest per-generation computational overhead, accelerated convergence (20–29% fewer generations, Table 5) yields net reductions of 30–37% in total computation time across all noise scenarios, with 35% savings achieved under baseline conditions (scenario S2, from 4.9 h to 3.2 h). This computational efficiency establishes practical viability for near-real-time digital twin updates in operational bridge monitoring systems.

5. Conclusions

This paper presents an enhanced NSGA-III framework for multi-objective parameter identification in cable-stayed bridge digital twins, addressing critical challenges in solution diversity maintenance, convergence efficiency, and measurement noise robustness. The main findings are summarized as follows:

- A unified multi-objective formulation balancing modal frequency matching, mode shape correlation, and static deflection error is established under physics-based constraints. Sobol global sensitivity analysis reduces the parameter space from 156 to 16 variables (89.7% reduction) while retaining 99% of system sensitivity, effectively addressing the curse of dimensionality.
- Three coordinated innovations improve standard NSGA-III: adaptive reference point relocation (every 20 generations) preserves diversity across non-convex Pareto regions; sensitivity-informed variation operators emphasize high-impact parameters; and intermittent quasi-Newton local search accelerates convergence by 25–30% while eliminating Pareto front gaps.
- On a 580 m-span finite element model, the enhanced algorithm reduces average identification error from 2.7% (standard NSGA-III) to

1.3%, increases hypervolume by 22%, and decreases total runtime by 35% despite 9.5% per-generation overhead. Identified parameters reproduce synthetic measurements with frequency correlation > 0.98, MAC > 0.95, and static residuals within $\pm 2\%$.

- Under realistic noise levels (1–5% frequency error, 0.5–2 mm deflection noise), average parameter errors remain below 3.0%, convergence accelerates by 20–29%, and the noise sensitivity coefficient of 0.75 (vs. 0.85 for standard NSGA-III) demonstrates graceful degradation suitable for field applications.

Current validation relies on synthetic data with controlled noise profiles; real-world measurements exhibit non-Gaussian noise, intermittent data loss, and unmodeled effects (soil-structure interaction, temperature gradients, traffic-induced vibrations) that may challenge identification accuracy. Furthermore, the framework performs offline calibration, whereas online adaptive updating would enable continuous model fidelity during operation. Future research should: (1) validate the approach on instrumented bridges with long-term monitoring data; (2) investigate surrogate-assisted evolutionary algorithms or reduced-order modeling to accelerate ultrahigh-fidelity simulations; (3) incorporate Bayesian uncertainty quantification for probabilistic parameter bounds; and (4) extend the methodology to suspension bridges, arch bridges, and building structures.

CRediT authorship contribution statement

Hongye Gou: Writing – review & editing, Validation, Supervision, Methodology. **Wenhao Li:** Writing – review & editing, Software, Methodology, Conceptualization. **Xin Huang:** Writing – review & editing, Visualization, Data curation. **Zhuang Tan:** Writing – original draft, Software, Methodology, Conceptualization. **Alireza Entezami:** Writing – review & editing, Validation, Conceptualization. **Junming Wang:** Writing – review & editing, Resources, Project administration. **Yi Bao:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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